Fiscal Year:	FY 2022	**Task Last Updated:**	FY 05/05/2022
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Project Title:	Harnessing Extremophilic Black Yeasts to Understand the Adaptive Strategies of Eukaryotes to Mars-Like Conditions		
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Program/Discipline:			
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Comments:			
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
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Flight Program:	Balloon Program Office		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Co-Investigators (Co-Is) Drs. Zachary Schultzhaus and Jillian Romsdahl left the Naval Research Laboratory (NRL) in 2021. Two new postdocs Drs. Jennifer Yuzon and Tiffany Wong-Stack joined the team as Co-Investigators. Dr. Yuzon's expertise is fungal biology and bioinformatics and Dr. Wong-Stack's expertise is plant biology and molecular biology.		
COI Name (Institution):	Yuzon, Jennifer (National Research Council/Naval Research Laboratory) Wong-Stack, Tiffany (American Society for Engineering Education/Naval Research Laboratory)		
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Task Description:	Fungi colonize nearly every ecosystem, including manmade structures. Species that produce airborne spores (i.e., molds) have a ubiquitous global distribution and are therefore the most commonly isolated. Yeasts, the premier model organisms for cell biology, are more restricted in distribution and diversity. Embodying characteristics of both are the black (melanized) fungi; they are hardy and thrive in a wide range of environments, but they do not produce airborne spores, so they can be studied in a restricted laboratory environment. Through natural adaptation to harsh conditions, black fungi have developed tools that situate them to both benefit (as bioremediators of hazardous waste and producers of valuable biomolecules) and threaten (as causes of disease and corrosion of materials) spaceflight and other ventures. These characteristics have motivated increased study of melanized yeasts in the last decade. We propose, here, to further develop these organisms for understanding microbial adaptation in the extreme conditions of space. This project will use state-of-the-art genetic tools to understand how spaceflight affects the survival strategies of black fungi to the conditions experienced on high-altitude Antarctic balloon flights, using species that are rationally chosen based on evolutionary relationships. The objectives are to: 1) understand how high-altitude affects survival of desiccated black yeast populations; 2) identify signatures of spaceflight on biomolecules (DNA, protein) in these organisms; and 3) analyze the effect of high altitude on the recovery of desiccated black yeast cultures, in order to identify processes related to the survival and adaptation of higher organisms (eukaryotes) to these environments. Biological samples will be produced in strictly resource-limited conditions to simulate the relevant, natural growth environments of melanized yeasts. Therefore, while the completion of this project will greatly increase our understanding of how fungi adapt to space-like conditions, an absolute minimal use of personnel and equipment time will be involved.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Antarctic Balloon piggyback opportunity will fly above the Antarctic and experience Mars-analog conditions, with a unique radiological profile. It also allows for a passive biological cargo to be flown as long as it maintains viability over an extended time through intense UV and ionizing radiation. To take advantage of this mission and produce the most valuable biological data possible, we are using the fungus Aspergillus niger to examine how two important cellular mechanisms, melanization and DNA repair (conserved throughout higher organisms) can assist in survival, protection, and adaptation to environments similar to Mars. This opportunity will yield results that greatly contribute to our knowledge of establishing life in Mars but also poses logistical hurdles that we will address in this report.
	Background The engineering challenges that must be overcome to send humans to Mars are daunting, but another formidable barrier to such a mission is the reality that the environmental conditions that await there are not compatible with the maintenance of life as we know it. The surface of Mars and the Moon are characterized by extreme temperature fluctuations, minimal atmospheric pressure, an absence of water and nutrients, and, perhaps most insidiously, enormous levels of radiation, both ionizing (IR) and non-ionizing (UV). Since space exploration began, however, the idea that certain organisms could survive unadulterated exposure to space or space-like conditions has maintained a moderate level of interest in the space biology community. Certain locations on Earth (rock surfaces in tundra and desert environments, for example) resemble the conditions seen on Mars, and life, though it does not necessarily flourish, is not excluded from these regions. Additionally, experimental evidence collected from space biology missions has identified some organisms, mostly lichens and melanized fungi, that, if processed correctly, can exhibit survival after exposure to outer space for up to 18 month. Finally, resistance to high doses of acute IR is currently being observed as a common trait of many fungi, and some of these organisms, particularly those that produce melanin, have been observed thriving in sites where levels of IR are high. The Antarctic Balloon piggyback opportunity will fly above the Antarctic and experience Mars-analog conditions, with a unique radiological profile. It also allows for a passive biological cargo to be flown as long as it maintains viability over an extended time through intense UV and ionizing radiation. To take advantage of this mission and produce the most valuable biological data possible, we are using the fungus Aspergillus niger (A. niger) to examine how two important cellular mechanisms, melanization and DNA repair (conserved throughout higher organisms) can assist in survival, protection, and adaptation to environments similar to Mars. This opportunity will yield results that greatly contribute to our knowledge of establishing life in Mars but also poses logistical hurdles that we will address in this report. Overview of the project Mars is characterized by high levels of UV and cosmic ionizing radiation (IR), desiccation, low temperatures, and low pressure. Fungi such as A. niger have recently been used as models for studying the biological effects of cosmic IR, with a focus on repairing DNA damage using mechanisms similar to that of animals. When double strand breaks in the DNA occur, fungi and animals use two mechanisms: 1) non-homologous end-joining (NHEJ), a rapid process which can theoretically fuse any two pieces of DNA together, and 2) homologous recombination (HR), which makes use of homologous chromosomes as templates to accurately achieve repair and reconstitution of a double strand break. In addition to maintaining the integrity of their DNA from radiation insult, fungi such as A. niger produce melanin, which is currently being explored for its ability to protect against detrimental effects of IR. Using a NASA flight to conduct our experiment, we will observe how DNA repair mechanisms and melanin affect the survival and adaptation of A. niger. In addition to a wild type (WT) strain, three mutant strains will be used in the experimental design: kusA, deficient in NHEJ; uvsC, deficient in HR; and fwnA, deficient in melanin production. These three mutant strains of A. niger have already been developed. All four A. niger strains will be integrated into the balloon using the Microbes in Atmosphere for Radiation, Survival and Biological Outcomes Experiment (MARSBOx) hardware. Each strain will be grown on solid Yeast Extract-Peptone-Dextrose (YPD) medium for 5-7 days until spores are ready to be harvested. At this point, spores will be washed and adjusted to a concentration of 1 x 109 cells/mL in sterile water. An aliquot of 200 µL of spore suspension will be evenly dispersed onto sterile quartz discs (6 cm in diameter), 16 per strain. The experiment will have a combined sample size of 64 cultures. Finally, the discs will be transferred to a vacuum desiccator to obtain dried samples for integration into the MARSBOx. Upon recovery of the samples, each disk will be rehydrated by placing each in 2 mL of sterile water and plated onto solid YPD medium. After incubating cells for 48 hours at 30oC, observations of individual spores will be used to measure:

	• Cellular survival (determined by the number of germinated vs. ungerminated spores) • Mutation rates (e.g., the distribution, type, and number of mutations induced due to cosmic radiation) • The space-adapted transcriptomic signature (i.e., the gene expression differences occurring in spores after exposure to Mars conditions) • Proteomic analysis (identity and function of proteins produced by tissue responding to a Mars-analog environment) With these measurements, we will be testing four hypotheses. First, we hypothesize that the disruption of melanin biosynthesis or DNA repair pathways will contribute to a decrease in survival under the conditions experienced by the Antarctic balloon flight. The second is that disrupting one or both of the DNA repair pathways will result in higher Structural Variant (SV) mutations (1kb or larger in size) occurring in the offspring of the mutant strains (kusA and uvsC) as a result of IR exposure. Genome sequencing will allow us to identify mutations resulting from deficiencies in both DNA repair pathways. Third, we hypothesize that adaption to Mars conditions will cause changes in gene expression, that we will be able to detect using transcriptomic analyses. Additionally, we will be able to identify proteins (e.g., ribosomes and oxidoreductases) responsible for recovery after exposure to Mars conditions using proteomics.
Task Progress:	In order to make the experiment a success, two main obstacles need to be overcome: 1) long-term survival during 12 months of desiccation, and 2) exposure to a wide range of temperatures. How we address these issues are as follows: • Long-term desiccation of our study organism is required for this experiment because the journey of the samples to the Antarctic and their return to the U.S. Naval Research Laboratory (NRL) can take up to 12 months. The payload will first travel by water to New Zealand and will then be flown to McMurdo Station in Antarctica. Exposure to UV and ionizing radiation at ~39km above the Antarctic surface will last a minimum of three months. A. niger is a suitable organism for this journey because it has a 30% survival rate after 15 months of desiccation according to previous studies in the Atacama Desert. • Resilience to a wide range of temperatures is also a challenge. During this time, samples will be exposed to ambient temperatures that will include freezing temperatures on the surface of the Antarctic as the payload waits to be loaded onto the gondola. Once the balloon reaches high altitude (~39km above the surface), the samples will be exposed to extremely warm temperatures (max. 55°C). Though the high temperature is beyond that of Mars, the engineering challenge of maintaining Mars temperatures is too great a challenge. Initially, the team intended to shield the samples from higher temperatures. However, this would reduce UV exposure and expose the samples only to low dose ionizing radiation. Previous findings have shown that short exposure to low doses of ionizing radiation in other Mars-analog environments have negligible effect on A. niger compared to UV radiation exposure. Therefore, the team has determined that UV exposure is necessary for Martian conditions and for informative results. To accomplish this, the experiment will have two sets of samples on the Antarctic balloon: 1) one set exposed to UV radiation, and as a result, high temperatures (max. 55°C); and 2) one set shielded from high temperatures and UV radiation, but still exposed to low doses of ionizing radiation. In addition, the NRL team will carry out survival assays of A. niger at 55°C which is below but close to temperatures (above 60°C) in which the fungus is killed. Year 1 goals and progress towards these goals Pilot Experiments: In Year 1 of the Antarctic project, our research group focused on survival assays of three candidate microorganisms: A. niger, Exophiala dermatitidis, and Exophiala lecanii-corni. This involved desiccating fungal tissue for several months; after which, the discs were rehydrated and survival rates were quantified. • Long-term desiccation and low survival rates: For each of the candidate species, two strains (melanized and unmelanized) were desiccated on filter paper and stored for one to three months. Afterwards, two discs per strain were rehydrated and plated onto YPD medium to count the number of viable vs. dead cells. A. niger (melanized: 91.1 ± 12.4% and unmelanized: 100.0 ± 0.0%) had the highest and most consistent survival rates after the first month, followed by E. lecanii-corni (melanized: 1.1 ± 1.6% and unmelanized: 3.5 ± 2.0%) and E. dermatitidis (melanized: 0.3 ± 0.4% and unmelanized: 0.7 ± 1.0%), respectively. In the third month, the survival rate was reduced for all species with 0.0% survival rate for all E. dermatitidis, and the melanized E. lecanii-corni strain. Therefore, we chose A. niger (three month survival rate for melanized strains: 11.2 ± 9.0% and unmelanized strains: 10.1 ± 5.3%) as the best candidate for the Antarctic balloon flight. Since previous reports have shown at least a 30% survival rate after 15 months of desiccation, the science team intends to modify the current protocol to increase the recovery of viable spores. We will follow the protocol of the Atacama Desert study by using quartz discs instead of filter paper so that fungal material can easily dislodge from the disc. Also, our previous protocol included a “metabolic slowdown phase” in which discs with fungal material were placed on minimal media before desiccation with the intention of gradually slowing down the metabolism of the fungus. However, this is likely counterintuitive for the spore-forming A. niger. Spores of A. niger are already in an inactive state and are durable to harsh conditions. Growth on the minimal media reactivates growth of the fungus so that it is no longer in the hardy spore phase. Therefore, we will remove the “metabolic slowdown phase” of our protocol to maintain A. niger in spore form. • Extreme fluctuating temperatures: So far the survival assays have been conducted at room temperature (~ 20°C). Once we have established a protocol that increases the survival rate of A. niger, we will repeat the survival assay using a similar temperature profile that mirrors the experience of going to, during, and coming back from the Antarctic balloon flight. • Assembling a team with expertise in space biology and engineering: The Antarctic balloon flight poses several challenges that require expertise from different groups. The NRL team provides the microbiology, radiation, and bioinformatics skills required to address the physiological and molecular changes as the fungus adapts to Mars-like conditions. The NASA Kennedy Space Center (KSC) team has the engineering background and spaceflight experience to ensure that the samples are exposed to an environment similar to Mars. The DLR team has extensive experience with microbiology experiments in space and will be able to help with setting up the payload. [Ed. Note: The German Space Agency (Deutsches Zentrum für Luft- und Raumfahrt) is abbreviated as DLR.] • Flight application: The team has already submitted a flight application for a long-term flight (min. 3 months) above the Antarctic.
Bibliography Type:	Description: (Last Updated: 06/06/2023)

Articles in Peer-reviewed Journals	Romsdahl J, Schultzhaus Z, Cuomo CA, Dong H, Abeyratne-Perera H, Hervey WJ, Wang Z. "Phenotypic characterization and comparative genomics of the melanin-producing yeast Exophiala lecanii-corni reveals a distinct stress tolerance profile and reduced ribosomal genetic content." J Fungi (Basel). 2021 Dec 15;7(12):1078. https://doi.org/10.3390/jof7121078 ; PMID: 34947060; PMCID: PMC8709033 , Dec-2021